Replication Guide

Technology Overload? Macroeconomic Implications of
Accelerated Obsolescence

Seda Basihos

Overview

This package contains all code and data required to reproduce the results in *"Technology Overload? Macroeconomic Implications of Accelerated Obsolescence."* The materials are organized into two top-level folders:

- `Simulations/` — model code, model calibration, simulation inputs and outputs.
- `Data and Figures/` — data construction, empirical estimation, and figure/table generation.

Software requirements. MATLAB, Stata, and R.

Folder: `Simulation/`

`Simulation/Calibration and Simulation/`

This subfolder contains the MATLAB files that calibrate the model and simulate its outcomes. The driver script is `RUN.m`. It endogenously determines the calibrated parameters by matching moments given the set of fixed parameters, and it also issues the simulation command (`main.m`). To generate the model outcomes (Table 1, Figure 2, and Figure 3), simply run `RUN.m`.

The remaining files do not need to be run directly, but are documented here for anyone who wishes to change the configuration or parameter values.

- `main.m` runs the simulation.
- `object.m` contains the moments to be matched and the set parameters. Anyone who wants to change the parameter values or the moment conditions can edit this file, together with `parini.m`.
- `ODE.m` is the transformed system of differential equations.
- `parini.m` sets initial parameter values.
- `StatEq.m` contains the (static) equilibrium conditions.
- `shock.m` stages the magnitude of the shock to technological obsolescence.
- `simulation.m` arranges the simulation results.
- `plot_.m` plots the summary of the transition and the data versus model comparison.
- The remaining `.m` files are system files of the relaxation algorithm of Trimborn et al. (2008).

Simulation/Robustness and Sensitivity/

This subfolder creates the sensitivity analysis results (Figure 4 and Figure 5). For Figure 4, run plot_sensitivity.m. For Figure 5, run separately plot_monopoly_distortions_bl_bk.m, plot_monopoly_distortions_L.m, and plot_monopoly_distortions_g_sigma.m. These respectively give different values of b_z and b_h (transformation parameters in the innovation sectors), L_h and L_z (resource allocation in the innovation sectors), and static changes in productivity growth and the labor share under the baseline obsolescence change given different combinations of parameter values.

This subfolder also includes params_comparetive.xlsb, which provides comparative statics (changes in productivity growth and the labor share) given different parameter values defined under Section 6.1 in the manuscript. The subfolder also contains Monopoly_distortions_sensitivity.R, an R script that calculates the parameter values b_h, b_z, L_h, and L_z for different combinations of α and β (the substitutability parameters for monopoly distortions).

Folder: Data and Figures/

Data and Figures/Obsolescence Rate/

This subfolder contains ARESULT_e_s.dta, which includes the capital stock, investment, and depreciation (all in both constant dollars and current dollars), and the service lives from the BEA Fixed Assets Accounts for the asset types, together with price deflators. The data contain several price deflators, but the analysis uses one of them, which is already set in the code. To replicate the results and calculate the physical depreciation rate, run construction.do, which also creates Figure 1. To obtain the obsolescence rate, run Figure_1_Table_2.do, which gives the Figure 1 and Table 2 results. Note that the Table 2 results appear in the Stata output window rather than as a separate table to be exported.

Data and Figures/CES Parameter/

This subfolder includes the data and codes for the estimation of the CES parameter under directed technical change. Run CES_paramer_estimation.do to obtain the Figure Ax.1, Table Ax.1, and Table Ax.2 results in Online Appendix A.

Data and Figures/Target Moments and Appendix/

This subfolder includes the macro data (Macro Data.xlsx, see the README sheet there for the data sources) for the long-run economic conditions of the US economy, which are used in the calibration of the model. Run target_moments_and_appendix.do. It first gives the moments in Table 1 (displayed in the Stata output window) and then Figure Bx.2, Figure Bx.3, and Figure Bx.4 in Online Appendix B.

Data and Figures/Cross Country/

This subfolder includes data on the labor share measures of 16 advanced economies (from the OECD ANA, OECD, 2019, and Penn World Table 10.01, Feenstra et al., 2015) and cumulative labor productivity growth (Fernald et al., 2025). Run cross_country.do to generate Figure Bx.5 in Online Appendix B.

Mapping: Paper Exhibits to Code

Exhibit	Script	Notes
Table 1 (parameters)	`RUN.m`	Make `main` a comment in `RUN.m` and check the MATLAB workspace, which gives the calibrated parameters using `fsolve`.
Table 1 (moments)	`target_moments_and_appendix.do`	Check the Stata output window.
Table 2	`Figure_1_Table_2.do`	
Table 3	`target_moments_and_appendix.do`; `RUN.m`	Initial BGP entries are the same as the Table 1 moment conditions. For new BGP entries (model), check the first two columns in `DATA_StSt.mat` at the workspace after running `RUN.m`.
Figure 1	`construction.do`, `Figure_1_Table_2.do`	
Figure 2	`RUN.m`	`RUN.m` produces this figure in MATLAB's default style. To reproduce the exact formatting used in the paper, run `plot.do` under the `Calibration and Simulation/` folder.
Figure 3	`RUN.m`	
Figure 4	`plot_sensitivity.m`	
Figure 5	`plot_monopoly_distortions_bl_bk.m`, `plot_monopoly_distortions_L.m`, `plot_monopoly_distortions_g_sigma.m`	
Table Ax.1	`CES_paramer_estimation.do`	
Table Ax.2	`CES_paramer_estimation.do`	
Figure Ax.1	`CES_paramer_estimation.do`	
Table Bx.1	`RUN.m`	The convergence speed and eigenvalue calculation appear in the MATLAB command window. For alternative parameters, manually change the parameter values in `object.m`, `parini.m`, and `shock.m`.
Figure Bx.1	`construction.do`	Obtained by uncommenting the line that drops the computing equipment and software asset types (`EP1A`–`EP1H`, `ENS1`–`ENS3`). This gives the obsolescence rate excluding computing equipment and software.
Figure Bx.2	`target_moments_and_appendix.do`	
Figure Bx.3	`target_moments_and_appendix.do`	
Figure Bx.4	`target_moments_and_appendix.do`	
Figure Bx.5	`cross_country.do`	

Notes

In the MATLAB and R scripts, the equation definitions use `b_k` in place of b_z and `b_l` in place of b_h. These are simply alternative labels for the same parameters, and the naming was left unchanged in the code to avoid unnecessary edits.

References

Feenstra, R. C., Inklaar, R., and Timmer, M. P. (2015). The Next Generation of the Penn World Table. *American Economic Review*, 105(10):3150–3182.

Fernald, J., Inklaar, R., and Ruzic, D. (2025). The Productivity Slowdown in Advanced Economies: Common Shocks or Common Trends? *Review of Income and Wealth*, 71(1):e12690.

Organisation for Economic Co-operation and Development (OECD) (2019). Annual National Accounts. https://www.oecd.org/en/publications/serials/national-accounts-of-oecd-countries_g1g12db2.html. Accessed: 2025-01-02.

Trimborn, T., Koch, K.-J., and Steger, T. M. (2008). Multidimensional Transitional Dynamics: A Simple Numerical Procedure. *Macroeconomic Dynamics*, 12(3):301–319.